Liam H Nguyen

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EDUCATION

Northeastern University – Boston, MA **Expected May 2028**
B.S. in Electrical and Computer Engineering | **GPA**: 3.75
Study Abroad: Northeastern University – London, UK **Sep 2024 – May 2025**
Honors: Dean's List (Fall 2024, Spring 2025, Fall 2025)
Relevant Classes: Calculus 1, 2, and 3, Circuits and Signals, Physics 1 and 2, Fundamentals of Networks, Linear Algebra, Differential Equations, Fundamentals of Electronics, Embedded Design: Enabling Robotics, Computing Fundamentals
Activities: Northeastern University Symphony Orchestra and Northeastern University Chamber Orchestra (Violin)

TECHNICAL SKILLS

Hardware: Oscilloscope, Function Generator, FPGA, Circuit Analysis and Design, Arduino, Raspberry Pi, I2C, Linux
Programming: Python, C++, NumPy, MATLAB
Tools: AutoCAD, SolidWorks, LTspice, PSpice, Quartus Prime

ENGINEERING PROJECTS

Multi-Sensor Thermal Imaging Platform – Personal | Raspberry Pi, AutoCAD, I2C **Feb 2026 – Present**

- Configured port forwarding and SSH remote access for external network management
- Designing 3D printed housing in AutoCAD for field utilization of the complete system
- Interfacing MLX90640 infrared thermal array, VL53L7CX ToF distance sensor, and BME280 environmental sensor over I2C on a Raspberry Pi 5 to enable thermal observation of circuit components

Binary Coded Decimal Converter – Northeastern | Quartus Prime, FPGA **Jan 2026**

- Designed and minimized combinational logic to decode 4-bit BCD inputs into seven-segment display outputs using truth tables and Karnaugh maps
- Implemented and verified combinational logic circuit through Quartus Prime simulations and programmed FPGA board to interface hardware switch inputs with seven-segment display outputs

DNS Sinkhole – Personal | Raspberry Pi **Dec 2025**

- Constructed a Raspberry Pi-based DNS Sinkhole to intercept DNS queries of unwanted domains
- Assigned a static IP address for the Raspberry Pi and configured the DNS settings of the router's DHCP server to utilize the DNS sinkhole at the router level

ECG Signal Filter and Optimizer – Northeastern | MATLAB, Circuit Analysis and Design **Nov 2025 – Dec 2025**

- Designed and built an analog ECG circuit using an instrumentation amplifier and high/low pass filters
- Utilized circuit analysis to tune cutoff frequencies and gain to improve signal clarity and further suppress noise
- Created a MATLAB program to digitize and digitally filter the signal

Sustainable Automatic Plant-Care System – Northeastern | Arduino, SolidWorks **Mar 2025 – Apr 2025**

- Designed and built a solar-powered automatic plant monitoring and care system using SolidWorks
- Installed a watering system and an environmental sensor display, used Arduino to program their response to stimuli from humidity, sunlight, and temperature sensors

WORK EXPERIENCE

Iridium Satellite – Leesburg, VA | Satellite Operations Trainee **May 2024 – Jun 2024**

- Systematized files to improve internal efficiency and structure
- Shadowed the Satellite and Network Operations team, developing an understanding of real-time troubleshooting of satellite operations
- Observed the Network Infrastructure team, gaining exposure to satellite network optimization and maintenance

Fort Washington Estates – Fort Washington, PA | Music Program Volunteer and Organizer **Mar 2024 – May 2024**

- Arranged and performed 4 concerts for a retirement community
- Coordinated schedule with Activities Director, selected and rehearsed repertoire